

Dynamic Diffuse Global Illumination Resampling

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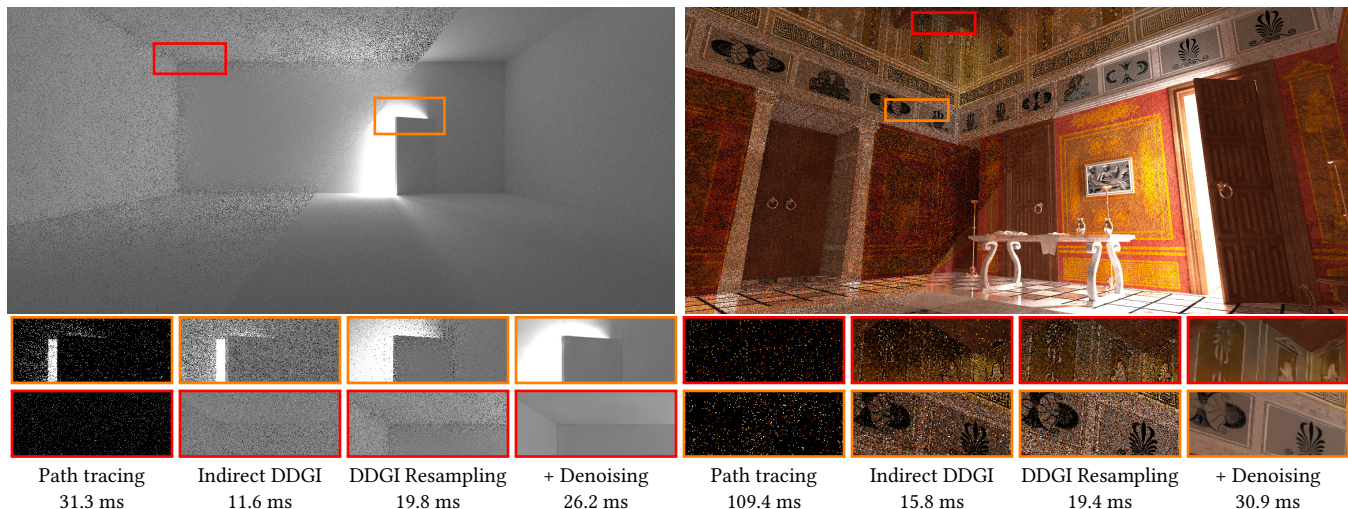


Figure 1: The large images show a split between our DDGI Resampling (left) and a path traced reference (right). The insets compare the noise levels of one sample per pixel (spp) path tracing, direct illumination by ReSTIR with indirect illumination gathered from DDGI at secondary vertices, and our DDGI Resampling. The final column displays our technique in combination with the OptiX denoiser. The path traced reference was generated offline at 32k samples per pixel, whereas our technique renders a 1080p frame in under 31ms on an NVIDIA RTX 3090, including the DDGI update and denoising.

ABSTRACT

Ray traced global illumination can be partitioned into direct contributions of the light sources that reflect to the camera after one bounce and indirect contributions that scatter for multiple bounces. We propose a new real-time solution called dynamic diffuse global illumination resampling that computes direct and indirect illumination accurately and with low noise. The key idea is to derive a new, unified algorithm from the principles of the state of the art ReSTIR many-lights direct shadowing [Bitterli et al. 2020] and the DDGI indirect light probes [Majercik et al. 2019] real-time algorithms. By this unification, global illumination resampling achieves higher quality than the combination of its two components at real-time framerates. At the cost of little bias, our technique also outperforms hardware accelerated path tracing in both runtime and noise.

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1 INTRODUCTION

In offline rendering, global illumination is computed by path tracing: for each pixel in the image, trace many stochastic multi-bounce paths through the image plane and average their contribution for the final result. This result can be noisy, so noise reduction techniques such as path guiding are often used [Vorba et al. 2019].

Though path tracing has gotten much faster recently, state of the art variants (at acceptable noise levels) are still far from the 60 frames per second target of modern games. Because of this, games have to resort to algorithms approximating global illumination. We present a new real-time global illumination algorithm that efficiently combines resampling with an irradiance probe volume.

2 ALGORITHM

In order to render global illumination in real-time, we combine spatio-temporal resampling [Bitterli et al. 2020] with an augmented variant of dynamic diffuse global illumination (DDGI) [Majercik et al. 2019] and a decomposition of the radiance into two integrals of bounded variance [Kollig and Keller 2004]. As illustrated in Figure 2, the phases of our algorithm are:

Sample the emissive surfaces at one sample per pixel and schedule these candidates for next event estimation.

Trace a secondary ray from the point to be shaded according to the non-specular part of the BRDF, convert its contribution to the area measure, and add it to the candidates for next event estimation. Shade x from the point of intersection y_s using the correction factor

$$\frac{\max\{G(x, y_s) - b, 0\}}{G(x, y_s)}. \quad (1)$$

Add the specular contribution by following the reflected ray.

Spatio-temporally resample a point y from the candidates [Bitterli et al. 2020] proportional to the sum of the emissive contribution and the irradiance from the DDGI volume with the geometry term $\min\{G(x, y), b\}$ bounded by a threshold $b \in \mathbb{R}^+$.

Trace a shadow ray to y and shade, if visible, using both the emissive contribution and the irradiance from the DDGI volume with the same bounded geometry term.

Bounding the geometry term bounds the variance in case the distance $\|y - x\|_2$ is small (we use $b = 1$). The resulting bias is compensated by the contribution sampled according to the non-specular part of the BRDF scaled by the factor in Eqn. (1) as derived by Kollig and Keller [2004], which amounts to a simple variant of multiple importance sampling [Veach and Guibas 1995] that combines well with ReSTIR.

Gathering radiance from the scene surfaces by the secondary rays requires DDGI to include direct illumination, which is a simple extension of Majercik et al. [2019] that can be achieved by importance sampling of emissive surfaces during the DDGI update step. Using the augmented DDGI approximation reduces variance as compared to using a photon map as a virtual point light cloud [Kollig and Keller 2004].

3 RESULTS AND DISCUSSION

Our algorithm runs at below 31ms per frame including denoising on an NVIDIA RTX 3090. We show results for our approach in Figure 1. The insets from left to right show unidirectional path tracing, ReSTIR for direct illumination and DDGI queried (without resampling) at the secondary vertex, our new DDGI Resampling, and the denoised result. Our method produces the lowest noise levels at interactive frame rates, at the cost of minimal bias as compared to the path traced reference. This allows for real-time denoising of complex scenes using our algorithm. The left image is a challenging case for our method because of the single light source and simple materials. Still, our method produces a result close to the path traced reference. On a scene with more complex illumination and materials (right image) our algorithm performs

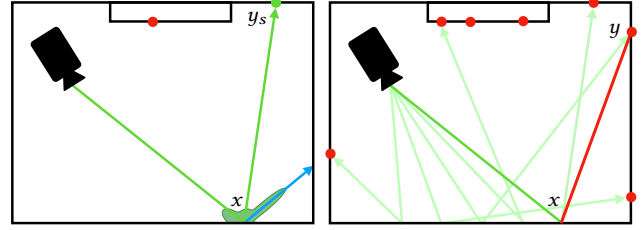


Figure 2: Our algorithm computes direct and indirect illumination in a unified way. Left: Points (red) are sampled on the emissive surfaces. Additional samples on the scene surface (green) are created by sampling the non-specular part of the BRDF at the primary vertex. Specular contributions are added by following the reflected ray (blue). Right: With all candidates across pixels converted to the area measure, the primary vertex extends a shadow ray (red) to a selected candidate. Its geometry term may be bounded to avoid over-modulation by the weak singularity, the bias of which is corrected by scaling the contribution of the non-specular BRDF sample that compensates for the bounding bias.

even better, as the medium frequency artifacts of the denoiser are masked by texturing and glossy/specular reflection.

Overall, our unified method:

- (1) Provides the benefits of resampling to direct *and* indirect illumination at the primary hitpoint, as opposed to only resampling the direct.
- (2) Improves the visibility artifacts of a limited probe grid resolution by using one bounce of ray-traced visibility before querying the irradiance volume.
- (3) Yields approximate indirect glossy reflections by allowing diffuse irradiance to contribute along primary ray glossy reflections.

4 CONCLUSION

Global illumination resampling achieves accurate light transport simulation in real-time with low noise. Including DDGI in the unified resampling process recovers much of the bias that would otherwise be inherent in the straightforward combination of resampling and irradiance caching schemes.

REFERENCES

- Benedikt Bitterli, Chris Wyman, Matt Pharr, Peter Shirley, Aaron Lefohn, and Wojciech Jarosz. 2020. Spatiotemporal Reservoir Resampling for Real-time Ray Tracing with Dynamic Direct Lighting. *ACM Transactions on Graphics (Proceedings of SIGGRAPH)* 39, 4, Article 148 (July 2020), 17 pages.
- Thomas Kollig and Alexander Keller. 2004. Illumination in the Presence of Weak Singularities. In *Monte Carlo and Quasi-Monte Carlo Methods 2004*, Denis Talay and Harald Niederreiter (Eds.). Springer, 245–257.
- Zander Majercik, Jean-Philippe Guertin, Derek Nowrouzezahrai, and Morgan McGuire. 2019. Dynamic Diffuse Global Illumination with Ray-Traced Irradiance Fields. *Journal of Computer Graphics Techniques (JCGT)* 8, 2 (5 June 2019), 1–30.
- Eric Veach and Leonidas J. Guibas. 1995. Optimally Combining Sampling Techniques for Monte Carlo Rendering. In *Proceedings of the 22nd Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '95)*. 419–428.
- Jiří Vorba, Johannes Hanika, Sebastian Herholz, Thomas Müller, Jaroslav Krivánek, and Alexander Keller. 2019. Path Guiding in Production. In *ACM SIGGRAPH 2019 Courses (SIGGRAPH '19)*. Article 18, 77 pages.